Electrical Safety Audit Report

of

HDFC BANK

Talwandi Bhai,

Punjab.

Branch Code: 1424 **Branch Type:** Semi urban

Executed By:

M/s APlus Automations

SWiSH SAFE-T
TESTING, INSPECTION – ELECTRICAL & ENERGY

www.aplusautomations.com

on

13th March, 2026

Report Summary & Key Findings

S/No.	Key Observations	Recommended Rectification Measures	Area
1.	Appropriate space not available.*(Ref Image11)*	Optimize UPS room space be removing unnecessary items.	UPS
2.	Door broken and missing in Main Panel.*(Ref Image 9,10)*	Replace the broken main panel door and ensure its properly secured with functional locks to maintain safety.	Main Electrical Panel
3.	Insulation mat not available in main electrical panel Room.*(Ref Image 9)*	Insulation mat should be provided to avoid any electrical incidents	Main Electrical panel
4.	Meter and Y phase indicator is not working. *(Ref Image 10)*	Should be replaced.	Main Electrical Panel.

1. General Inspection, Comments & Recommendation

S/No.	Parameters need to be checked	Observation	Remarks
1	If any foreign material is kept nearby & top of the panel & inform store manager for its removal?	Yes	Extra material found in Server Room and Panel Room.
2	Check Basic housekeeping and hygiene for electrical system i.e. Electrical Panels, DBs, ACs, DGs, Servo-Stabilizer, Inverter, UPS, Batteries etc.	Not Cleaned	Housekeeping in UPS Room should be maintained.
3	Ensure Basic Housekeeping and tidiness of general area for any untidy storage in the rooms, passages, corridors, balconies, Pantry rooms, washrooms, server rooms, ATM rooms etc.	Not Cleaned	Mains Meter: Need to clean and also the UPS Room need to Clean
4	Ensure Basic Housekeeping and tidiness of the outside area for unwanted and untidy storage of fire-prone material like empty carton boxes, wooden material, cotton waste, thermocol, general waste, overflowing dust bins, damaged plastic waste, damaged furniture, paper waste, used oil, fuel cans etc.	Cleaned	Outside area is clean
5	Material around or above LT Panel & Equipment?	Yes	Carton boxes near Panel observed.
6	Material Around DG?	No	No Extra material was found near DG
7	Diesel/Oil/Coolant spillages observed around DG?	No	DG Canopy is rusted
8	Any oil and fuel stains around DG and on the canopy?	No	DG Canopy is rusted
9	Are canopy doors – locked, not locked, open, damaged etc.?	DG: Locked	DG: Locked
10	Is DG set clean from outside?	Yes	DG is Clean
11	Is material stacked inside of canopy?	DG: No	No Stacking Observed
12	Are all emergency numbers displayed and easily visible?	No	Emergency numbers not displayed.
13	Is all type of sign-board displayed?	Yes	Available
14	Are HUB rooms or electrical rooms locked?	UPS: Yes	UPS: Locked
15	Check electricity bill with all the aspects like Name of power supply company, type of power supply, category, multiplying factor, TOD, penalties, average consumption in terms of Unit & Rs etc.	Yes	Ref. Section 12 (Meter Detail)
16	Note all connection details like LT/HT connection, connected load, sanctioned load etc.	LT	Ref. Section 12 (Meter Detail)

2. Power Distribution & Electrical Installation

S/No.	Parameters need to be checked	Observation	Remarks
1	Availability of SLD & Proper nomenclature in PDB's /LDB /UPSDB	No	Not Available
2	Is SLD and actual installation matching?	No	Not installed
3	Is Electrical Cable's size as per design?	Yes	Meet the load requirements
4	Is glanding & termination of all incoming & outgoing power cables done properly?	Yes	Mains EI Panel :OK
5	Is any Hanging Cable in entire premise?	No	No hanging cable observed
6	Is the IR / ER Test carried out?	No	Not Done (Requires Shutdown)
7	Check all protective relays, MCBs, MCCBs, SFUs, Fuses, RCCB, ELCB etc. are installed and are healthy?	Yes	MCCB, MCB & Contactors installed
8	Check Setting of Protective relay?	Yes	Found OK
9	Check all the switch controls and handles/knobs functionality	Yes	Available
10	Check changeover rating for DG	Yes	200A
11	Check changeover rating for Servo bypass	No	Not Visible
12	Check voltage at the changeover terminal of DG, if okay?	DG: Yes	Ref. Section 13 (Supply Detail)
13	Check the voltage RYB and RN, YN, BN	Yes	Ref. Section 13 (Supply Detail)
14	Check the current in Main Panel & all Distribution	Yes	Ref. Section 13 (Supply Detail)
15	Check condition of electrical safety rubber mats & ISO Standard	No	Not Provided
16	Is an exhaust fan available in the electrical panel room?	No	AC Installed.
17	Is Load balancing is proper in Incoming & all major distributions?	Yes	Load is balanced.
18	Is one terminal having only one wire?	Yes	Terminal Connections Ok
19	Any decolorization of any of the power cable terminations?	No	Decolorization not observed.
20	Is double earthing for all the panels and equipment?	No	Double Earthing is not provided
21	Is there any foreign and unwanted material around the panel?	No	Panel area is clean
22	Are all openings of electrical panels closed and vermin proof?	No	Main Panel door missing.
23	Are all indication lamps, meters, selector switches are in good and working condition?	No	EI Panel: Indicator & Meter not working

24	Note down the reading of power factor capacitors, their switch gears, APFC controller	NA	Not available
25	Check capacitor current and identify if any capacitor is weak or not taking current	NA	Not available
26	Is EB meter for proper visibility and seal?	Yes	Seal Intact
27	Note down the all required EB meter details	Yes	Ref. Section 12 (Meter Detail)
28	Is there any unauthorized connection / tapping on the meter?	No	No un authorized connection observed.
29	Observe any overloading at any distribution point?	No	There is no overloading
30	Inspect all the sockets, switches etc. in the area	Yes	Working
31	Is any socket connected with a spike guard and the socket is getting overloaded?	No	No spike guard observed
32	Is the power supply for signage and lollipops is proper?	Yes	Working
33	Shock treatment chart availability	No	Shock chart not available
34	Is proper conduiting used for wiring on the False ceiling?	Yes	Available
35	Check any wires below the carpet, if the floor is carpeted	No	Floor is not carpeted
36	Are any rodent traces, dirt, water etc.in cable raceway?	No	No infestation observed
37	Is there proper ferruling and nomenclature on the panels, cables, switch gears?	No	Ferruling not available
38	Are any wall openings at cable entry and exit?	Yes	Cable entry in server room not closed.
39	Whether identification between UPS and Raw supply in the sockets is done?	UPS: Yes	Raw Power & UPS Power identification available
40	Is Housekeeping staff aware about this identification between UPS and Row supply in the sockets?	UPS: Yes	Response team is Informed

3. Review of Protective Earthing System

S/No.	Parameters need to be checked	Observation	Remarks
1	List down and check condition of earth pits	No	Earth Pits not visible
2	Are multiple earth pits available in premises for electrical system?	No	Earth Pits not visible
3	Check whether earthing grid is form, if yes check individual and over all resistance	No	Earth Pits not visible
4	Measure the earth pit resistance	Not Done	Earthing Pit not found
5	Confirm load balancing, by checking the neutral current	Yes	Load is balanced
6	Note down voltage between earth and neutral	1.7V	Within Range ≤ 2V.

4. Review of Operation & Maintenance System

S/No.	Parameters need to be checked	Observation	Remarks
1	Is PPM and Break down records of the equipment is maintained?	No	Not maintained ppm
2	List down of all the equipment like ACs, DG, Servo-Stabilizer, Inverter, UPS, Transformer etc.	Listed	Ref. Section 15 (Equipment Details)
3	Is condition & health is proper of ACs, DG, Servo-Stabilizer, Inverter, UPS, Transformer?	No	Diesel Generator (DG), UPS
4	Note down running electrical parameters including ratings of ACs, DG, Servo Stabilizer, Inverter, UPS, Transformer	Yes	Ref. Section 15 (Equipment Details)
5	Is isolation (LOTO) / permit process followed by technicians?	No	No technician available
6	Note the type of lamps CFL, LED etc.	Noted	Ref. Section 14 (Load Details)
7	If all lights are working, if not make report for the same	Yes	Working
8	Check LUX level at various locations in the customer area, BOH, pantry, washroom, corridors, locker room etc.	Yes	Ref. Section 17 (LUX Levels Report)
9	Is washroom lights and equipment like hand dryer, exhaust, motion sensor (if installed) etc. are in working condition?	Yes	Working

5. Emergency Power/ Lighting System

S/No.	Parameters need to be checked	Observation	Remarks
1	Check whether emergency power is supplied either by Inverter or DG	Yes	UPS
2	Is DG operation is proper by switching off the mains supply with due permission from site in-charge?	DG: Yes	DG: Checked
3	Are emergency lights in working condition by switching off lighting supply?	No	Checked

6. CCTV, Access & Control for Electrical Installations

S/No.	Parameters need to be checked	Observation	Remarks
1	Are all Access control points are working?	Yes	Working
2	For ensuring continuous functioning of security alarm, fire alarm and CCTV system, whether the power supply is provided through the UPS?	Yes	Ups
3	Is access control system integrated with the fire system?	Yes	System is Integrated
4	Did Lift, Air conditioning system switches off with the activation of fire alarm system	No	Not Activate
5	Check the MCP availability	Yes	Available

7. Lightning Protection System

S/No.	Parameters need to be checked	Observation	Remarks
1	Check lightning arrester?	Yes	Not available
2	Is an earth pit for lightning arrester is proper?	No	Not available
3	Is getting proper continuity in lightning arrestor?	No	Not available
4	Overall condition of the system	Yes	Not available

8. Fire Protection

S/No.	Parameters need to be checked	Observation	Remarks
1	Are fire extinguishers being at designated place, visible and in good condition?	Yes	Fire Extinguisher placed in good condition
2	Is the tag of inspection & its validity is available on Fire Extinguisher?	Yes	Fire Extinguishers are healthy.
3	Is the FAS panel healthy?	Yes	Healthy
4	Are there any active faults / troubles / alarms on the display?	No	No fault
5	Are all Smoke / heat detectors are not capped? If no, inform admin manager and remove the same	Yes	Capped
6	Check emergency evacuation plan is displayed at all relevant locations	No	Not available
7	Check whether First aid medicines and equipment are available	Yes	Available

9. Staff Competency

S/No.	Parameters need to be checked	Observation	Remarks
1	Is FA (Facility Attendant), HK (House-Keeping) enough competent on the electrical safety and action in case of electrical emergency? (Confirm by interview)	Yes	Emergency response team is well informed
2	Check Competency of any branch staff with the permission from Branch Manager on electrical safety, hazards and actions required. (Confirm by interview)	Yes	Checked. Emergency response team is well informed

10. Thermography

S/No.	Parameters need to be checked	Observation	Remarks
1	Separate report of thermography including normal and thermal images with reason of hotspot and recommendation	Separate report of inspection	*Ref. Thermography Report*

11. MoM with Branch Manager for additional concern of Electrical Safety Audit, if any:

HDFC BANK

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MoM with Branch Manager for additional concern of Electrical Safety Audit if Any:

1. SLD N/A Main Panel.
2. DG Not maintained.
3. critical Area server room and Panel room Extra material need to remove.
4. Shock chart N/A critical Area need to Paste.
5. electricity Bill Not Provide –
6. Main Panel Door Broken and Missing
7. 2 Sign board not working

Auditor Name, Contact & Signature: Amit Kumar, 8529865920, Branch code. – 1424

Branch Manager Name, Contact & Signature:

Electrical Safety Audit Agency Name: Tirender Preet singh 9356415612

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12. Meter Details

Check Parameters	Reading / Observations	Recommendations / Remarks
Type of Metering (LT/HT)	LT	
Electrical Power Input	415V	
Category	Commercial	
Condition of transformer neutral earthing in case same is dedicated for our Bank	No	Transformer is not dedicated.
Sanctioned Load	NA	Electrical bill is not provided
Connected Load	35.62 kWh	*Ref. Section 14 (Load Detail Report)*
Penalty due to excessive loading during last quarter (if connected load/maximum demand in a month is more than sanctioned load/contract demand)	No	
Average power factor during last quarter	NA	Electrical bill is not provided
Penalty due to low power factor	NA	Electrical bill is not provided
Average Billing/Month	NA	Electrical bill is not provided

13. Supply Detail Report

Mains Meter - Power Supply Details - 3 in 3 out							
Voltage				Current		Earthing	
RN	232	RY	436	R	6.4	RE	235
YN	236	YB	435	Y	11.4	YE	237
BN	237	BR	439	B	3.7	BE	236
				N	7.9	NE	1.4

Mains Incomer - Power Supply Details - 3 in 3 out							
Voltage				Current		Earthing	
RN	236	RY	427	R	7.2	RE	233
YN	239	YB	429	Y	4.7	YE	228
BN	241	BR	425	B	12.0	BE	225
				N	9.3	NE	1.2

Diesel Generator (DG) - Power Supply Details - 3 in 3 out							
Voltage				Current		Earthing	
RN	227	RY	415	R	12.2	RE	114
YN	226	YB	414	Y	7.6	YE	116
BN	227	BR	415	B	11.4	BE	98
				N	4.2	NE	1.1

Mains Electrical Panel - Power Supply Details - 3 in 3 out							
Voltage				Current		Earthing	
RN	223	RY	428	R	7.2	RE	223
YN	228	YB	426	Y	4.7	YE	226
BN	224	BR	423	B	13.6	BE	225
				N	9.3	NE	1.7

UPS - Power Supply Details - 1 in 1 out							
Voltage				Output			
PN	230	P	2.1	PN	221	P	2.8
NE	0.8	N	2.0	NE	1.1	N	2.6
PE	229			PE	220		

14. Load Detail Report

Description	Quantity	Load required to run single equipment KW	Total power/Load requirement (KW/KVA)	Remarks
Air Conditioner - Split	4	1.5	7.20 kWh	Working
Air Conditioner - Package	2	8.5	17.00 kWh	Working
ATM	2	1000	2.00 kWh	Working
CCTV	18	15	0.27 kWh	Working
Currency Counter	1	500	0.50 kWh	Working
Currency Sorter	1	300	0.30 kWh	Working
Exhaust Fan	3	50	0.15 kWh	Working
LED Panel	18	24	0.43 kWh	Panel light
Other Lamps	34	18	0.61 kWh	Working
PC/Laptop	20	70	1.40 kWh	Laptop
PC/Laptop	8	250	2.00 kWh	Pc
Printer	10	200	2.00 kWh	Working
Refrigerator	1	100	0.10 kWh	Working
Signboard	2	100	0.20 kWh	Not working
RO Filter	1	65	0.07 kWh	Working
UPS	1	15kVA	12.00 kWh	Single phase ups
UV Lamp	7	15	0.10 kWh	2 lamp not working
Lolipop	3	50	0.15 kWh	Not working
Wall fan	9	70	0.63 kWh	Working
Round light	34	15	0.51 kWh	Working
Total			**35.62 kWh**	Approx. Excluding UPS

15. Equipment Details

Description	Check Parameters	General Check	Recommendations / Remarks
DG Set Details	Number of DG installed	1	Available
	DG Capacity and Make	Yes	DG: - Kirloskar
	DG body and neutral earthing, if ok?	No	DG: Not available
	Is DG maintained properly?	DG: No	DG: Need to maintain
	DG cable specifications	DG: Yes	DG: 70Sqmm
	Condition of DG Cable Insulation & Termination Visually	DG: Weak	DG: Dg not proper maintained
Servo Details	Number of Servo Installed	1	Available
	Servo Capacity and Make	75kVA	Ganpati Electricals
	Is the stabilizer body earthed?	No	Should be Earthed
	Is stabilizer Bypass switch installed?	Yes	Available
	Are Amp & Volt meters working at servo stabilizer	No	Not working
	Condition of Servo Cable Insulation & Termination Visually	Yes	Satisfactory
UPS Details	Number of UPS Installed	1	Available
	UPS Capacity and Make	Yes	UPS:15kva Make :3EM
	Rating of incomer of UPS output DB (MCB/Isolator/ELCB/RCCB)	UPS: 63A	Available
	Is earth leakage protection there in the UPS output DB	UPS: OK	UPS: OK
	Status of UPS (Online/Bypass/Faulty)	UPS: Online	UPS: Working
	Are battery racks provided with earthing?	No	-
	Batteries	Yes	-
	Make	Yes	Amaron
	Quantity	15	-
	AH Rating	UPS: 26	UPS: Working
	Are the batteries having any sign of Bulging or any other deformity	No	Healthy
	Availability of Battery Rack Cover	UPS: No	Not available

16. Earthing Details

Check Area	Observation	Recommendations / Remarks
Earthing Type (Chemical / Conventional)	Chemical	Not found earth pit
Number of Earth Pit(s)	NA	Not found earth pit
Condition of Earth Pit(s)	NA	Not found earth pit
Material used for Earthing (Wire/Strip)	Wire	OK
Voltage of Earthing	1.7V	Neutral EARTH
Resistance of Earthing	Not Done (Requires Shutdown)	Earth pit Not found

17. LUX Levels Report

Check Area	Observation	Recommendations / Remarks
Customer Area	378	Ok
BOH Area	319	Ok
Pantry	116	Ok
Washroom	164	Ok
Corridors	321	Ok
Locker Room	222	Ok

18. Photographs

Mains Meter	
Image 1	Image2
Need to clean	Seal

Servo Stabilizer	
Image 3	Image 4
Servo Stabilizer	Meters not working

Diesel Generator (DG)

Image 5

DG not properly maintained

Image 6

Cobwebs inside DG Panel

Image 7

DG Panel

Mains Electrical Panel

Image 8

Infestation traces observed

Image 9

Insulation Mat not Provided

Image 10

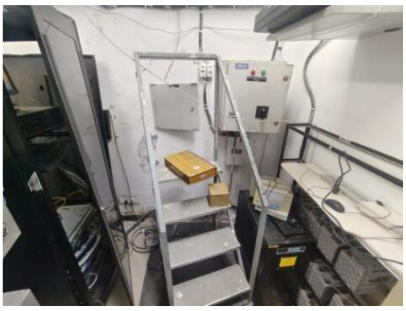
Panel door broken and missing
Indicator & Meter not working.

UPS

Image11

Ups room need to maintain

Image12



Space limited in server room

General Inspection

Image13

Lolipop and signage are not working

19. Thermal Inspection Report:

THERMOGRAPHY	
S.No	**Description/ Observations**
1.	**Thermal Imager Specifications:** The Specifications of the thermal imager used for the said electrical thermography are given below. 1. Type: FLIR TG267 2. Temperature Range: -20° to 380°C 3. Detector type: 180X120 True Thermal Image
2.	**Introduction:** Thermography is a term used to describe a type of photography that uses infrared radiated wavelengths to make pictures as opposed to visible light as in normal photographs. It can be also referred to as 'thermal imaging' or 'infrared'. Objects that have a temperature above absolute zero (-273.15°C or 0 Kelvin) emit infrared wavelengths. Thermography is the production of thermal (heat) pictures from these wavelengths, whereby temperature measurements or comparative analysis can be made. This analysis was commissioned to establish the temperature condition of the electrical installations. In particular, the analysis was to investigate any evidence of hot spot, unbalanced load etc. Where possible, images were to be captured in a manner such that all possible areas could be presented to the customer in a clearly recognizable format. A short narrative would accompany the images to highlight any salient points. This survey and report combines images to provide a general thermographic overview of the electrical installation, together with a selection of close-up observations of areas with interest. Whilst care has been taken to record temperatures as accurately as possible, the absolute values obtained should be treated with a degree of caution. Variable environmental conditions together with changes in camera angle, distance, material change and emissivity can all adversely affect the result. However, by combining images with equal parameters the 'relative' change of temperature across a selected material will be accurate and therefore useful analysis can be made. From discussions prior to the survey, the following points have been incorporated into the methodology of data collection and presentation of findings. 1. The survey to be a combination of general scanning and capturing of selected images that may fairly represent the situation. 2. Anomalies noted to be identified either within the narrative or on the image. 3. The infrared inspection was performed with the electrical installation in the “as found” condition. 4. Emissivity is set from the emissivity table of the known material, whereas reflectivity has been set manually for each location.

3. **Repair Note Rating :**

Each thermograph is given a Subjective Repair Priority Ratings, which is determined on risk of failure, based upon our opinion, of how critical the subject item is to the energy efficiency & safe and profitable operation of your overall system.

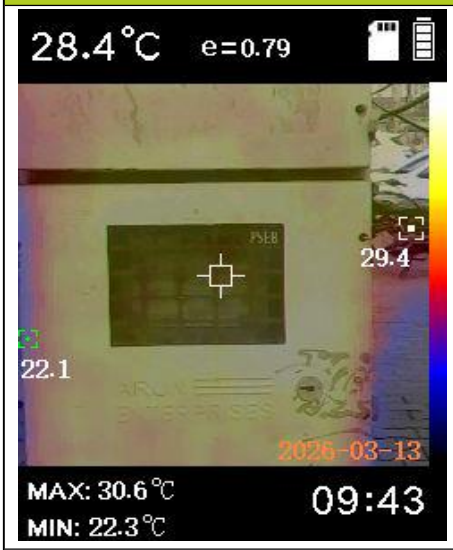
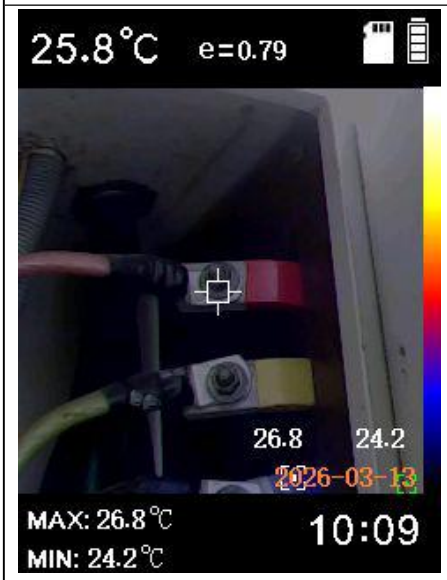
The Subjective Repair Priority considered in the report is expressed as under-

	Not Operational (NO) at the time of Infrared Inspection	
	NORMAL at the time of Infrared Inspection	
	Further Investigation required	
	Beginning of Problem	-Monitor Routinely
	Problem Exists	-Monitor Closely
	Serious Problem	-Attend at the Earliest Opportunity
	Critical Situation	-Repair Immediately

Temperature levels considered for Severity Priority for electrical items are as under-

Temperature Rise Above Ambient	Δ T*	Situation	Repair Priority
10^{0}C to 20 ^{0}C	1 ^{0}C to 3 ^{0}C	Beginning of Problem	Monitor Routinely
20°C to 40 ^{0}C	4 ^{0}C to 9 ^{0}C	First Stage of Overheating	Monitor Closely
40°C to 60 ^{0}C	10 ^{0}C to 14 ^{0}C	Second Stage Overheating	Monitor very Closely
60°C to 80 ^{0}C	15 ^{0}C to 19 ^{0}C	Excessive Overheating	Repair as soon as posible
Greater than 80 ^{0}C	➢ 20 ^{0}C	Acute Overheating	Repair Immediately

*Δ T between similar components under similar load

Identification	Thermo gram	Repair Priority
		Area: Mains Meter Amb. Temp = 28.4 °C Max. Temp. = 29.5 °C ΔT =1.1 °C **Risk Category**: Normal No hotspot observed
		Area: DG Amb. Temp = 27.7 °C Max. Temp. = 29.1 °C ΔT =1.4 °C **Risk Category**: Normal No hotspot observed
		Area: Mains Electric Panel Amb. Temp = 25.5 °C Max. Temp. = 27.6 °C ΔT = 2.1 °C **Risk Category**: Beginning of Problem **Recommendations:** Monitor Routinely.

5. **Summary Of Scanned Areas**

A table of the area scanned during this study, its location and anomaly status at the time of the infrared scan is given below. The actual thermographic images of the anomalies are given in the table. For all anomalies, we have included a control photograph identifying the equipment (regular photograph) and a thermographic image (thermal image) of the area where the anomaly was found.

We have scanned the Electrical Installation of your facility with an Infrared Imager. As per the severity priority rating, we have categorized the anomalies and their classification is given below:

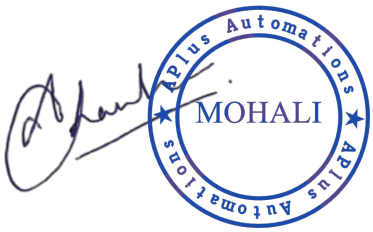
	Description	Nos.
	Not Operational	--
	Further Investigation required	--
	Normal	2
	Beginning of Problem	1
	Problem Exists – Monitor Further	--
	Serious Problem – Attend at the Earliest Opportunity	--
	Critical Situation – Repair Immediately	--
	Total	**3**

It may be noted that the temperature rise criteria specified in the report is as per the guide-lines of NFPA70B

20. Certification

I/We certify that the installation has been checked and tested by me/us and that to the best of my/our knowledge and belief, the overall outcome of audit was:

- [ ] Good
- [x] **Satisfactory**
- [ ] Requires Attention
- [ ] Critical



Stamp & Signature of Licensed Electrical Engineer: Arun Chauhaan
License No. and name of Issuing Authority - 50590000007720 (Govt. Of NCT of Delhi)